

A Constant Vector Beats Your Digital Twin: On the Evaluation of Cluster-Level LLM Personality Simulation

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Abstract

Large language models are increasingly prompted to act as “digital twins” of human personality groups, and such systems are routinely evaluated by how closely the model’s questionnaire answers match those of real respondents. We show that this evaluation is broken. For the answer-similarity metric used throughout this literature, we prove that a simulator which reproduces the target population’s response distribution *exactly* is strictly dominated by a constant vector, and that the gap is governed by the difference between the Gini mean difference and the mean absolute deviation about the median. On 410,168 IPIP-NEO-120 respondents this structural penalty is 7.75 points. Empirically, across five LLMs and four personality clusters, a per-item cluster mean scores 0.796 while the best evolutionary-optimised persona reaches 0.739, losing in 20 of 20 cells; a *real human drawn from the target cluster* scores 0.721 and also loses to the constant in 20 of 20 cells. We further find that the released benchmark scores raw model self-reports against reverse-recoded human answers on 55 of 120 items; we correct this and report every result under both orientations, the replacement suite included. A metric that ranks a constant above a genuine group member cannot be measuring personality simulation. We propose a replacement suite — a classifier two-sample test, per-item distributional fidelity, and a variance ratio — each reported against an explicit human ceiling and constant floor, and show it inverts the ranking correctly. Under it, current personas are near-perfectly separable from humans (C2ST-AUC 0.994 vs. a human ceiling of 0.564) and reach only 0.43 of human response *spread* as a standard-deviation ratio. We release code and per-cell results.

1 Introduction

Simulating human respondents with large language models has become a common instrument in computational social science and in LLM-based user

research. A widely used recipe conditions a model on a natural-language personality description and validates it by administering a psychometric inventory: the model answers the same Likert items a human answered, and the system is scored by the agreement between the two answer vectors.

This paper argues that the scoring step invalidates the conclusion. Our starting point is a concrete artefact. In a cluster-level digital-twin system built on IPIP-NEO-120 (Johnson, 2014), evolutionary prompt optimisation raises answer-level agreement while simultaneously *degrading* the recovered Big Five profile — mean absolute error, trait similarity and facet similarity all move the wrong way. That pattern is usually reported as an open puzzle. We show it is a predictable consequence of the metric.

Our contributions are:

1. **A proposition (§3)**: under answer-similarity, a simulator that faithfully samples the target population’s response distribution is *dominated* by the best constant predictor — strictly so on every IPIP-NEO item we measure. The penalty is the excess of the Gini mean difference over the mean absolute deviation about the median, divided by the scale range.
2. **An empirical audit (§4)**: across five LLMs $\times$ four clusters, on the original system’s own train/test splits, trivial constant baselines beat every model in every cell, and so does the global mean, which ignores the personality clusters entirely.
3. **A diagnosis of the human ceiling (§8)**: a real respondent from the target cluster — the ideal a cluster-level twin can aspire to — also loses to the constant in every cell.
4. **A replacement suite (§9)**: C2ST-AUC, distributional fidelity and variance ratio, each anchored by a human ceiling and a constant

floor. It ranks humans above constants, as any valid measure must.

5. **A cluster-validity audit** (§13): the $k=4$ partition these systems condition on is not supported by any internal criterion.

2 Setting

Task. A population of respondents is partitioned into K personality clusters by k -means over the 30 IPIP-NEO facet scores. For each cluster a natural-language description of the centroid is used to condition an LLM, which then answers the 120 IPIP-NEO items on a 1–5 Likert scale. Prompt wording is optimised by an evolutionary search whose fitness is answer agreement with held-out human members of the cluster.

The metric. A simulated answer vector x is scored against a human answer vector y by

$$S(x, y) = \frac{1}{J} \sum_{j=1}^J \left(1 - \frac{|x_j - y_j|}{4}\right), \quad (1)$$

with $J=120$ and the divisor 4 the scale range. Cell scores average S over the held-out respondents. Equation (1), or a monotone transform of it, is the objective the search optimises and the number reported.

3 A constant is optimal under S

Fix a cluster and an item j , and let P_j be the cluster’s response distribution over $\{1, \dots, 5\}$. Write $Y \sim P_j$ for a randomly drawn human respondent.

Proposition 1. *Let m_j be a median of P_j , let $\text{MAD}_j = \mathbb{E}_Y |m_j - Y|$ and let $\text{GMD}_j = \mathbb{E}_{X,Y} |X - Y|$ for $X \sim P_j$ independent of Y (the Gini mean difference). Then*

- (a) *the best constant predictor attains expected score $1 - \text{MAD}_j/4$;*
- (b) *a faithful simulator, i.e. one whose output is distributed as P_j , attains expected score $1 - \text{GMD}_j/4$;*
- (c) *$\text{GMD}_j \geq \text{MAD}_j$, with equality iff X lies almost surely in the median set of P_j .*

Hence the constant is at least as good as the faithful simulator, and strictly better unless P_j places all its mass on medians.

Proof. (a) and (b) are immediate from (1). For (c),

$$\begin{aligned} \text{GMD}_j &= \mathbb{E}_X [\mathbb{E}_Y |X - Y|] \\ &\geq \mathbb{E}_X [\min_c \mathbb{E}_Y |c - Y|] \\ &= \mathbb{E}_Y |m_j - Y| = \text{MAD}_j, \end{aligned}$$

since the median minimises expected absolute deviation. Equality requires X to attain that minimum almost surely, i.e. to lie in the median set of P_j . This is weaker than degeneracy: a balanced two-point law $P(Y=1) = P(Y=5) = \frac{1}{2}$ has $\text{GMD} = \text{MAD} = 2$, so the bound is tight there despite large variance. On the IPIP items it is strict throughout (Table 2). $\square$

The consequence is not a technicality. Under S , the ideal behaviour we actually want — reproducing the population’s response distribution — is penalised relative to emitting one fixed vector. Optimising S therefore pushes a simulator toward degenerate central-tendency responding, destroying exactly the answer variance that carries a personality profile. This is the mechanism behind the “answer similarity up, profile fidelity down” pattern reported as a puzzle in prior work.

3.1 The metric is CRPS with its dispersion term deleted

Proposition 1 is not an accident of Likert data. The continuous ranked probability score for a predictive distribution F and outcome y is

$$\text{CRPS}(F, y) = \mathbb{E}_F |X - y| - \frac{1}{2} \mathbb{E}_F |X - X'|,$$

and it is *strictly proper*: uniquely optimised by $F = P$. Equation (1) is its first term alone. Deleting the second is exactly what makes a constant optimal.

This yields the natural repair. Define

$$S_\lambda = 1 - [\mathbb{E} |X - y| - \lambda \mathbb{E} |X - X'|] / 4. \quad (2)$$

$\lambda=0$ is the audited metric — improper, constant-optimal. $\lambda=1/2$ is CRPS — strictly proper, truth-optimal. $\lambda>1/2$ is improper in the opposite direction. Crucially $\lambda=1/2$ is **not** a tunable hyperparameter but the unique proper choice, so the repair cannot be dismissed as a metric tuned to produce a preferred conclusion.

Sweeping λ over the same 20 cells (Table 1) shows the repair works and what it costs the simulator. At $\lambda=1/2$ the human ceiling (0.859) rises decisively above the constant floor (0.796): the

λ	Model	Constant	Human
0 (audited)	0.726	0.796	0.720
0.25	0.749	0.796	0.789
0.5 (CRPS, proper)	0.772	0.796	0.859
0.75	0.796	0.796	0.929
1.0	0.819	0.796	0.998

Table 1: The S_λ family of (2), averaged over 20 cells, all values on the corrected orientation. A constant wins at the audited $\lambda=0$; at the proper $\lambda=1/2$ a real cluster member wins, as it must. All-pairs scores, slightly below the paired ones of §4; the ordering is the same under either.

Clu.	n	MAD	GMD	S_{const}	S_{faith}
0	107,174	0.804	1.107	0.7990	0.7232
1	108,880	0.723	1.008	0.8192	0.7480
2	108,867	0.809	1.128	0.7978	0.7181
3	85,247	0.871	1.204	0.7822	0.6990
Mean		0.802	1.112	0.7996	0.7221

Table 2: Proposition 1 on 410,168 IPIP-NEO-120 respondents. $S(\text{const})$ is the best constant predictor; $S(\text{faithful})$ is a simulator that reproduces the cluster response distribution exactly. The faithful simulator is worse in every cluster.

ordering that §8 shows is inverted under the audited metric is *correct* under the proper one. Humans overtake the constant at $\lambda^*=0.30$, comfortably below the proper coefficient. The model needs $\lambda^*=0.75$ (median grid crossover; 0.757 on the aggregate curve) — a rule that over-rewards dispersion — and beats the constant in only 2 of 20 cells at the proper $\lambda=1/2$, though it does overtake in most cells once λ reaches 1, where the reward for dispersion is twice proper. So the simulator’s failure is not an artefact of a bad metric: it survives the correct one.

Table 2 evaluates the bound on the real data: the structural penalty against a faithful simulator averages **7.75 points**, and $\text{GMD}_j \geq \text{MAD}_j$ holds on 120/120 items in all four clusters.

4 Empirical audit

Setup. We re-score a published cluster-level digital-twin system on its own artefacts: the same 410,168-respondent IPIP-NEO-120 corpus, the same $k=4$ clusters, and the exact train/test respondent IDs it recorded, for five LLMs (GigaChat3-10B, GPT-4.1-mini, GPT-4.1-nano, Grok-4.1-fast, Qwen3-235B) $\times$ four clusters. We add no LLM calls: model answers are read from the released run artefacts. Baselines are fitted on the recorded train

IDs and scored on the recorded test IDs.

Baselines. *Cluster mean*: the per-item rounded mean of the cluster’s training respondents, one constant vector. *Cluster majority*: the per-item mode. *Global mean*: the per-item grand mean over all respondents, ignoring clusters. *Empirical sample*: sample each item independently from the cluster’s training response distribution. *Real human*: copy the answer vector of a randomly chosen training respondent from the same cluster.

Result. Table 3 reports S . The evolutionary persona averages 0.660 as released and 0.739 after the orientation correction of §7; the constant cluster mean averages 0.796. The LLM loses in 20 of 20 cells under either orientation, by 13.6 and 5.8 points respectively. What survives correction is *matched-cell* dominance: the model loses to its own cluster’s constant in every cell, while its best cell (0.807) now exceeds the weakest constant elsewhere (0.779), so the comparison only holds within a cluster. The global mean — which never looks at the cluster — scores 0.720.

The effect is below the protocol’s noise floor.

Re-drawing which 100 respondents are sampled, holding the 60/40 split ratio fixed, moves the *parameter-free* cluster-mean baseline by 1.3–2.5 points across 20 random draws. The improvement claimed for evolutionary optimisation in prior work is +0.72 points. With 40 test respondents per cell and a single fixed split, the reported effect is not separable from which rows landed in the test set.

5 What is a persona worth, in respondents?

“It loses to a constant” invites the reply that the constant is a strawman. So we restate the finding in a unit that is harder to dismiss: how many real respondents you would have to sample before a plug-in estimate of the cluster’s response distribution matches what the persona gives you.

Drawing m real respondents, forming the per-item empirical distribution, and sampling a synthetic population from it yields distributional fidelity 0.723 at $m=1$, 0.797 at $m=2$, 0.915 at $m=21$ and 0.923 at $m=34$. The curve is evaluated at the same sizes as the audited scores — forty synthetic against forty held-out humans — because DF depends on sample size, so a curve built at any other size cannot calibrate them. Reading the audited systems off it:

Model	Cluster				Mean
	0	1	2	3	
GPT-4.1-mini	0.657	0.638	0.710	0.652	0.664
GPT-4.1-nano	0.673	0.678	0.697	0.670	0.679
GigaChat3-10B	0.711	0.691	0.737	0.672	0.703
Grok-4.1-fast	0.598	0.583	0.659	0.614	0.613
Qwen3-235B	0.634	0.615	0.677	0.642	0.642
<i>Cluster mean (constant)</i>	0.794	0.827	0.785	0.779	0.796
<i>Real human, same cluster</i>	0.720	0.747	0.715	0.704	0.721

Table 3: Answer similarity S on the original system’s own test IDs, in the *original* orientation; correction adds 0.079 on average and changes no ordering. The constant (bold) beats every LLM in every cell; a real human scores below it.

System	DF	Resp.-equiv.
Evolutionary LLM persona	0.783	2
Constant cluster mean	0.796	2
Human ceiling	0.939	off-curve

An evolutionary-optimised persona is worth about **two** real respondents: its DF clears the one-respondent interval (0.783 against an upper bound of 0.766) and sits inside the two-respondent interval $[0.761, 0.825]$, where the constant also sits. The human ceiling is not a point on this curve at all — it is a direct sample of forty humans rather than a plug-in resample, and exceeds the curve even at $m=500$ — so we no longer express it in respondent units. This is the same result as §4 in different units, and it is the version we would put to a practitioner weighing synthetic respondents against a small pilot survey.

6 Does conditioning on the right cluster matter?

No persona study we know of reports the wrong-persona control, so no reported persona effect is currently attributable to the persona. We score every simulator against every cluster’s humans — 80 cells — and test diagonal dominance by permutation.

Metric	Matched	Mismatched	Δ
S	0.728	0.663	+0.065
DF	0.782	0.705	+0.077
C2ST	0.996	0.985	−0.010
VR	0.431	0.436	−0.005

Matched comparisons improve S and DF; they show no improvement in C2ST or VR. We report these gaps descriptively and attach no p -value: a permutation that shuffles the 80 cell labels freely destroys the matched structure of the five 4×4 matrices, and the models share clusters and personas, so its null is not the right one. Rescoring

fixed outputs against different human populations also does not isolate a causal effect of changing the prompt. This is the behavioural counterpart of §9 — the persona moves which items get endorsed, and not who is endorsing them — and it makes the wrong-persona control as necessary a baseline as the constant floor.

7 An orientation defect, and what survives it

Before comparing anything, the two sides must be on the same scale. They are not.

The released corpus stores human answers *reverse-recoded* so that a higher number always means more of the trait, while the simulator is shown the literal item text — “Cheat to get ahead” — and asked how accurately it describes it, i.e. it answers *raw*. On the 55 negatively-keyed items of the IPIP-NEO-120 the two scales run in opposite directions, and every published model-vs-human comparison is computed across that flip.

Two signatures make this unambiguous. Negatively-keyed items average 3.373 against 3.425 for positively-keyed ones — level, whereas raw self-reports on undesirable content sit far lower. And “Cheat to get ahead” carries a human mean of 4.15 out of 5.

The effect is large and it runs *against* the simulator. Correcting the orientation raises the correlation between the model’s and the humans’ per-item mean profiles from +0.09 — indistinguishable from no relationship at all — to +0.63, and raises S from 0.660 to 0.739.

What matters is which conclusions survive. Three do, and the invariances are stated before the numbers rather than after:

- **Proposition 1 is invariant.** MAD and GMD are computed on the human distribution alone,

and both are reflection-invariant.

- **The variance ratio is exactly invariant**, since $\text{sd}(6 - x) = \text{sd}(x)$. The under-dispersion finding of §9 is untouched.
- **The constant still wins every cell**. Reorientation closes the gap from 13.6 to 5.8 points, but the constant cluster mean beats the corrected model in 20 of 20 cells.

Correcting the defect in fact *sharpens* the paper’s claim. The corrected model scores 0.739, *above* the 0.721 of a real member of the target cluster — while a classifier separates that same model from real members at AUC 0.994 (§9). A metric that ranks a near-perfectly detectable simulator above the humans it fails to imitate is not measuring human-likeness under any reading.

We recommend an orientation check as a standing first step for benchmarks of this kind: correlate the simulator’s per-item mean profile against the population’s, and treat a near-zero correlation as a scale defect until shown otherwise.

8 The human ceiling sits below the constant floor

The decisive check is what a *real* member of the target cluster scores. A cluster-level digital twin cannot do better than being an actual member of the cluster; that is the definition of success.

A randomly drawn same-cluster respondent scores 0.721 — and is beaten by the constant cluster mean in 20 of 20 cells (Table 3, Figure 1, left). This is exactly Proposition 1 realised: a real human *is* a draw from P_j , so their expected score is $1 - \text{GMD}/4$, which the constant’s $1 - \text{MAD}/4$ must exceed. The predicted values (0.7996 constant, 0.7221 faithful) match the measured ones (0.796, 0.721) to within 0.4 points.

A measure that ranks a fixed vector above a genuine group member is not measuring group membership. Any ranking it induces over simulators — including the ranking prior work uses to select prompts — is therefore uninterpretable as evidence about personality fidelity.

9 A replacement suite

A usable metric must (i) be maximised by distributional fidelity rather than central tendency, and (ii) be reported against an explicit *human ceiling* and *constant floor*, so a number can be read as progress toward a real target rather than in the abstract. We

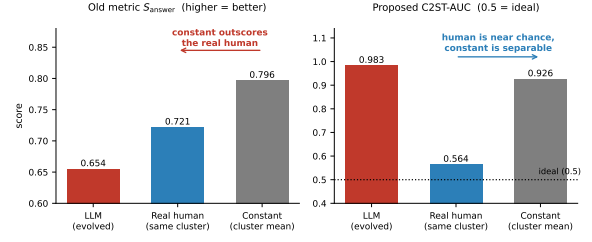


Figure 1: Left: under the old metric the constant outcores a real human. Right: under the proposed C2ST-AUC the human sits near chance (0.5, ideal) and the constant is near-perfectly separable. Only the right panel orders the three correctly.

propose three, computed from a set of simulated respondents rather than from paired individuals.

C2ST-AUC (primary). A classifier two-sample test: train a classifier to separate n simulated answer vectors from n held-out human vectors of the same cluster, and report cross-validated AUC. 0.5 means indistinguishable — the ideal twin. A constant is trivially separable. This directly operationalises the twin claim, and unlike S it cannot be gamed by collapsing onto the mean.

Distributional fidelity. $1 - \frac{1}{J} \sum_j W_1(\hat{P}_j^{\text{model}}, \hat{P}_j^{\text{human}})/4$, with W_1 the per-item Wasserstein-1 distance. Rewards matching the full response distribution.

Variance ratio. $\frac{1}{J} \sum_j \text{sd}(\text{model}_j)/\text{sd}(\text{human}_j)$; 1.0 is ideal, a constant gives 0. A diagnostic, not an optimisation target.

Results. Table 4 applies the suite to the same 20 cells. All three order human > constant, as required; the old metric does the reverse. Two substantive findings follow. First, current personas are *near-perfectly separable* from real humans (C2ST-AUC 0.994 against a human ceiling of 0.564) — and are in fact *more* distinguishable than a constant vector (0.926), because they miss both the location and the spread of the human distribution. Second, they are systematically under-dispersed: VR = 0.43, a ratio of standard deviations, so LLM personas produce well under half the response *spread* of the humans they simulate (Figure 2). Evolutionary optimisation moves none of these appreciably.

Under-dispersion is the wrong summary. A single variance ratio hides which variance is missing. Decomposing each answer matrix as $Y_{pj} = \mu + a_p + b_j + (ab)_{pj} + e$ — respondent main effect,

Metric	LLM	Human ceiling	Const. floor
C2ST-AUC ↓	0.994	0.564	0.926
DF ↑	0.783	0.939	0.795
VR (1.0 ideal)	0.431	1.018	0.000
$S_{\text{answer}} \uparrow$ (old)	0.737	0.721	0.796 †

Table 4: The suite over 20 cluster $\times$ model cells, each metric against its human ceiling and constant floor. †: under the old metric the constant floor *exceeds* the human ceiling — the failure this paper documents.

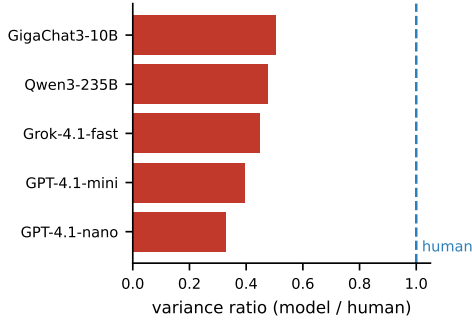


Figure 2: Response variance relative to the humans being simulated. Every model is substantially under-dispersed.

item main effect, and the person-by-item interaction that carries individuation — tells a sharper story. The simulator emits nearly the *right total* variance (ratio 0.942 over the 20 cells), but places it in the wrong component: the item main effect takes 81.2% of its variance against 29.8% for humans, while the interaction takes 17.7% against 67.0%. The interaction SD ratio is 0.466, and it is 0 for a constant by construction.

So the simulator is not emitting too little. It is emitting a confident cluster-level *answer key* — which items are endorsable — and almost none of which *person* endorses which item. We recommend reporting the interaction ratio rather than the plain variance ratio, since the two rank systems differently and only the former measures the quantity a personality simulator exists to reproduce.

9.1 What is C2ST actually reading?

A metric that separates simulators from humans at AUC 0.994 invites the objection that the classifier detects formatting rather than personality. A feature ladder (App. B) shows the objection has real force: five content-free response-style scalars already recover most of the separability above chance. It is not only style — ipsatising each respondent, or using facet scores alone, still separates

nearly as well — so style and content are redundant routes to the same conclusion. We therefore recommend reporting C2ST with its style-only rung alongside it, so readers can see how much of any separability is carried by response style. A simulator that closed only the style gap would look far better on the headline number while remaining just as distinguishable on content.

The anchors are estimated at $n=40$ per side. Appendix E sweeps n and shows the C2ST human ceiling is entirely finite-sample classifier bias (extrapolating to 0.500) while the DF ceiling is entirely plug-in bias (extrapolating to 1.000, i.e. pessimistic), and recommends $n \geq 40$ per side as a floor.

10 The proposition, reproduced in one experiment

Fitting on half of each cluster and scoring on the other half, two oracles differ only in what they predict: a **mean** oracle (ridge on the ordinal code), which predicts the conditional mean, and a **faithful** oracle, the same fit plus a resampled residual, which reproduces the conditional *distribution*. The mean oracle scores $S = 0.802$; the faithful one, strictly closer to the truth, scores $S = 0.730$ — worse by 7.2 points, against the 7.75 that the GMD — MAD gap predicts. Predicting the distribution correctly is penalised, and by very nearly the predicted amount. This construction does *not* settle whether the prompt limits the simulator: it reads a richer code than the ten scores the belief prompts carry, and restoring marginal spread by resampling residuals is not evidence of faithful conditioning. App. D takes that question up on the actual channel.

One further number bounds what conditioning buys at all: the mean oracle beats the constant (0.792) by 0.9 points. A perfect reader of the persona gains under a point over ignoring it.

11 Does this generalise beyond one questionnaire?

Every result so far comes from IPIP-NEO-120, so the objection is that the audit describes that questionnaire rather than the task. We re-test the two *structural* findings — the ones needing no LLM — on an independent instrument, construct and population: the Short Dark Triad (SD3), 17,740 respondents on 27 items.

Both replicate. $GMD \geq MAD$ holds on 27/27 items, with a structural penalty of 8.67 points

against IPIP’s 7.75; and the per-item constant again outscores a real respondent from the same population (0.744 vs 0.664) in 200 of 200 bootstraps, matching the inversion of §8. The defect belongs to the metric applied to ordinal questionnaire data, not to the Big Five or to one corpus.

12 Where the variance is lost

Knowing the simulator is mis-dispersed does not say *where* the dispersion goes. We read the model’s own belief before any sampling occurs, on open-weight models served locally.

Two stages. For each persona and item we take the model’s next-token distribution over the answer tokens $\{1, \dots, 5\}$ directly from the logprobs, then sample each item from its own belief and score the resulting population. One implementation detail matters: hybrid reasoning models emit a `<think>` token first — with probability ≈ 1 — so a naive readout sees no mass on the scale at all; we disable thinking and prefill the assistant turn.

Result. What is collapsed is the *individuating* part of the belief, not the belief’s total spread. In the 2024–25 models both are small; in the 2026 reasoning-first models the total spread reaches or exceeds the human marginal while individuation stays as low as ever.

The statistic matters. The quantity comparable to a human *across-respondent* SD is not the average within-persona conditional SD but the SD of the persona-*mixture* distribution, from the law of total variance $\text{Var}(X_j) = \mathbb{E}_i[\text{Var}(X_j | i)] + \text{Var}_i(\mathbb{E}[X_j | i])$, computed exactly from the retained probability vectors. The within-persona term alone would understate a faithful simulator, which can hold sharp per-persona beliefs and still match human variance through differing persona means.

Model	Params	VR _{with}	VR _{betw}	VR _{tot}
Qwen2.5-7B-Instruct	7B	0.367	0.193	0.430
Qwen2.5-32B-Instruct	32B	0.242	0.220	0.343
Qwen2.5-72B-Instruct	72B	0.367	0.295	0.488
Mistral-Small-24B	24B	0.538	0.177	0.579
QwQ-32B (reasoning)	32B	0.729	0.154	0.753
Qwen3-235B-A22B-2507	235B	0.695	0.258	0.758
Qwen3.5-9B	9B	1.058	0.167	1.078
GLM-4.7-Flash (MoE)	29B	1.209	0.265	1.242
gpt-oss-20b (MoE)	21B	1.121	0.126	1.131
Granite-4.1-8B	8B	0.386	0.207	0.457
Apertus-8B-Instruct	8B	0.927	0.130	0.939
<i>human</i>	—	—	—	1.000

Essentially all probability mass lands on the five valid answer tokens (0.97–1.00), so no model

has difficulty *emitting* a number. Belief entropy nonetheless spans 0.42–1.38 nats (maximum $\ln 5 = 1.609$), and we can offer no clean account of it: it tracks neither release date (Granite-4.1, 0.49) nor reasoning-tuning (QwQ 0.97; the standard-instruct Apertus 1.24). The 235B model is the one the audited system used, so this measures that system, not a proxy.

Total dispersion and individuation are decoupled. The decomposition says something the total does not, and the evidence for this is existential rather than correlational. gpt-oss-20b and Qwen2.5-72B invert cleanly: the former has $2.3\times$ the total dispersion of the latter (1.131 vs 0.488) and *less than half* its individuation (0.126 vs 0.295). The rank correlation over the eleven models is $\rho = -0.30$, but its bootstrap 95% interval is $[-0.92, 0.42]$, so we rest nothing on it.

QwQ-32B sharpens it: its dispersion is almost all per-item uncertainty (0.729 of 0.753), and it individuates less (0.154) than far smaller models.

The Qwen2.5 family isolates this at fixed family and varying scale: individuation rises monotonically, $0.193 \rightarrow 0.220 \rightarrow 0.295$ at 7B, 32B, 72B, while the total does not, moving $0.430 \rightarrow 0.343 \rightarrow 0.488$.

One bound survives across the whole set — a $34\times$ parameter range, six families, three generations, dense and mixture-of-experts, and both standard and reasoning-tuned models: no model exceeds $\text{VR}_{\text{between}} \approx 0.30$. Individuation sits between an eighth and under a third of the human value throughout, and neither the largest model nor the most dispersed one is the best at it.

The prompt is not what caps individuation. A persona is individuated only through a five-word quantisation of ten numbers, but that channel carries more than the models extract: a supervised conditional-mean decoder fitted on held-out respondents reaches 0.482 on these same personas, where the best model delivers 0.295, and no two personas share a code (App. D). The decoder is a reference, not a bound — a mis-calibrated simulator could exceed it by dispersing wrongly — so it shows the conditioning carries signal.

How much of it points the right way. $\text{VR}_{\text{between}}$ measures how far apart a simulator places its persona means, not whether that spread identifies the right respondent. Correlating each model’s persona means against the oracle’s over the forty per-

sonas — both on the corpus’s recoded scale — gives $r = 0.16\text{--}0.52$. The aligned amplitude $A = \frac{1}{J} \sum_j r_j \text{sd}(M_j) / \text{sd}(H_j)$, in units of human item SD, is $0.033\text{--}0.183$, against 0.489 $[0.461, 0.517]$ for real respondents: the best model reaches about a third of that, the weakest under a tenth. Intervals resample persona rows jointly across model, decoder and respondents with the decoder fixed. A separates models $\text{VR}_{\text{between}}$ does not — GLM-4.7 and Qwen3-235B differ by 0.007 in spread but 0.017 in aligned amplitude.

A simulator can therefore look more dispersed simply by being less confident: VR_{total} confounds uncertainty with individuation, and $\text{VR}_{\text{between}}$ is what a persona claim should be judged on.

One limit, plainly: the strong form of the claim below — that no sampling-side remedy could help — needs total dispersion well under the human marginal, which is false for the largest 2024–25 model (0.758) and for the three that *exceed* it ($1.078, 1.131, 1.242$); there, spread is present and is lost afterwards. What no model supplies at any scale is *individuation*.

Three consequences follow. For the low-dispersion models the spread is not discarded at decoding; it was never present, so recovering it there means *adding* dispersion, not revealing it: $p^{1/T}$ lifts Qwen2.5-7B from 0.43 to 1.02 at $T=5$ while supplying none of the alignment A measures. Since all mass lands on valid answer tokens, accounts blaming mode collapse on difficulty *emitting* numbers are unsupported here. And it closes the loop with the answer-level results: the item main effect dominates realised variance (§9) and conditioning on the correct cluster does nothing for dispersion or distinguishability (§6) — three measurements agreeing that persona conditioning moves *which items* get endorsed, not *who* endorses them.

13 Is $k=4$ a partition at all?

The systems we audit condition on a $k=4$ k -means partition of the 30 facet scores. We re-derive it and test $k \in \{2, \dots, 15\}$ under three preprocessings. Our $k=4$ solution reproduces the published labels ($\text{ARI} = 0.946$), confirming the pipeline.

No criterion supports $k=4$. Silhouette falls monotonically from 0.134 at $k=2$ to 0.053 at $k=15$, taking 0.082 at $k=4$, and Davies–Bouldin and the gap statistic select no interior k . Bootstrap stability looks at first like a defence ($\text{ARI} 0.903\text{--}$

0.977 for $k \leq 5$), but structureless surrogates matched to the facet covariance reproduce it exactly — $\text{ARI} 0.911$ and 0.921 against the real data’s 0.910 — while a separated control confirms the test has power ($\text{ARI} 1.000$). At $n=410,168$ bootstrap agreement measures estimator variance, not multimodality. The clusters are not demographic artefacts either (Cramér’s V of $0.055\text{--}0.057$); they are a reproducible discretisation of a continuum, which bounds the cluster-conditional signal any persona can exploit (App. C). Since stability-based selection is common, the general point is worth stating: at large n , bootstrap agreement must be read against a covariance-matched null or it certifies nothing.

14 Related work

Personality conditioning of LLMs is commonly validated by administering psychometric inventories to the conditioned model (Serapio-García et al., 2023; Jiang et al., 2024), and prompt sensitivity of the resulting traits is well documented (Dong et al., 2025; Gupta et al., 2024). Individual-level simulation from interview transcripts achieves high behavioural fidelity (Park et al., 2024) at a per-person cost that motivates the cluster-level shortcut we study. Evolutionary and search-based prompt optimisation (Guo et al., 2024; Prasad et al., 2023; Zhou et al., 2022) supplies the optimiser. Our contribution is orthogonal to all of these: we show the objective they are scored against is maximised by a degenerate solution.

Classifier two-sample tests are standard for comparing distributions (Lopez-Paz and Oquab, 2017), and the failure mode we identify — an absolute-error objective minimised by a conditional central tendency — is why such losses blur predictions in generative modelling.

15 Conclusion

Cluster-level LLM personality simulation is evaluated with a metric that a constant vector wins and a real human loses. We proved why, measured the structural penalty at 7.75 points, confirmed the inversion in 20 of 20 matched cells, and proposed a suite that orders humans above constants. Beneath the metric the simulation is thinner still: individuation that identifies the right respondent reaches at best a third of the human reference. Score with $S_{1/2}$ and report the floor and ceiling beside every number.

Limitations

Our audit covers one corpus (IPIP-NEO-120), one clustering pipeline, one optimisation family, and five models; the belief-level measurement of §12 covers eleven open-weight models spanning 7B–235B, six families and three generations, and the cross-instrument replication of §11 covers only the two structural findings, since SD3 carries no simulator runs; since it requires logprob access the API models in the audit do not provide; the within-family trend rests on three points and the cross-family comparison on eleven, so both are directional rather than fitted laws; all models are read with thinking disabled and the assistant turn pre-filled, and the four least confident of them (belief entropy above 1 nat) should not have their above-human total dispersion read as calibrated uncertainty, though low confidence cannot inflate VR_{between} , which is computed from persona means; two elicitation details proved to matter more than we expected — reading these models through a chat-completions server returned almost no mass on the answer tokens where the forward pass recovers 0.99–1.00, and harmony-format models require the assistant turn to open a channel before the pre-fill, without which mass on the scale falls to 0.39 — so a null result in this literature may report a readout rather than a model; every model we report is unquantised at inference, one FP8 checkpoint having resisted measurement through both available routes (its serving path returns no mass on the scale, and the forward-pass path hits a quantiser bug), so we make no claim about how quantisation affects individuation; the proposition is general to the absolute-error metric in (1), but the empirical magnitudes are specific to this setting. The model answers we re-score were produced by prior runs we did not control, at temperatures we did not set, and one released run was committed with entirely missing answers and is excluded. Our human ceiling is a single same-cluster respondent, which underestimates what a well-specified individual-level twin could achieve; we make no claim about individual-level methods. C2ST-AUC depends on classifier capacity and on $n=40$ per side, so its absolute value should be read as a lower bound on separability, and §9.1 shows most of what it reads is response style; we fold the AUC as $\max(\text{AUC}, 1 - \text{AUC})$, which places the null above 0.5 (0.517 signed against 0.569 folded) and makes our small- n floor a property of this classifier rather

than of C2ST; and DF treats items independently and so ignores within-respondent answer dependence. The corpus is self-reported, skewed toward respondents aged 18–25 and toward the USA, UK and Canada, so neither the clusters nor the ceilings should be read as representative of any broader population. Finally, we show current personas fail this evaluation; we do not show that any method can pass it.

Ethics statement

We use a public, de-identified corpus of self-reported personality questionnaire responses and add no new human-subjects data collection. Our results are cautionary: they indicate that current cluster-level personality simulations are readily distinguishable from real respondents and under-represent human response variance, and so should not be used as substitutes for human participants in research, product, or policy decisions. Reporting an evaluation number without its constant floor risks overstating the fidelity of synthetic respondents, which is the specific harm this paper aims to reduce. Simulating demographic or personality groups also risks reinforcing stereotypes when cluster descriptions are treated as characterisations of real people rather than as statistical summaries.

A Reproducibility

Every analysis in this paper is deterministic given the released corpus and the committed run artefacts, and requires no LLM calls except the belief-level measurement of §12. We ran the full analysis pipeline independently on two clusters with different operating systems, Python versions and library versions (scikit-learn 1.2 vs 1.7, pandas 2.1 vs 3.0). Every reported figure — the λ^* values, the orientation deltas, the variance components, the null-calibration ARIs of §13, and the oracle ceilings — agreed to all printed digits. Code and per-cell results are released, including the full belief simplex for every model and encoding variant we report: (40 personas $\times$ 120 items $\times$ 5) per cluster, with the persona ordering fixed by the released corpus so every reported number can be recomputed from the artifacts alone. Runs that failed the validity guard ship too, marked invalid, so the record shows what was measured and rejected. One transform maps model answers onto the corpus’s recoded scale and is imported by every analysis that compares the two.

B The C2ST feature ladder

A metric that separates simulators from humans at AUC 0.994 invites an obvious objection: perhaps the classifier is detecting formatting rather than anything about personality. We test our own metric with a feature ladder, each rung scored identically against the same anchors. The ladder is computed on the *original* orientation throughout, so its rungs are comparable to each other but not to the corrected suite, and its facet rung re-reverses the already-recoded human answers; a corrected version must handle orientation feature-by-feature rather than by one global flip. We therefore read the ladder qualitatively.

Features	LLM	Human	Const.
Full 120 items	0.983	0.564	0.926
5 style scalars only	0.950	0.566	0.994
Items, ipsatised	0.975	0.566	0.986
30 facet scores only	0.987	0.581	0.944
Forward-keyed only	0.990	0.568	0.810
Reverse-keyed only	0.984	0.583	0.843

Five content-free scalars — per-respondent mean and standard deviation, extreme -response rate, longest identical run, and forward-versus-reverse consistency — already recover most of the separability above chance here. The objection has real force, and we state it plainly: **a large part of what C2ST detects is response style**. We quote no share, since the ladder’s orientation handling is not yet consistent enough to support one.

It is not only style. Ipsatising each respondent (removing their own mean and standard deviation, i.e. the two largest style features) still leaves AUC 0.975, and facet scores alone give 0.987. Style and content are redundant routes to the same conclusion rather than one masking the other. The human ceiling is also stable at 0.56–0.58 across every rung, so the anchor does not move with the feature set.

C Cluster validity in full

No *separation* criterion supports $k=4$. Silhouette falls monotonically from 0.134 at $k=2$ to 0.053 at $k=15$, taking 0.082 at $k=4$; Davies–Bouldin is also minimised at $k=2$; the gap statistic increases monotonically and selects no interior k . A silhouette below 0.1 indicates essentially no separated structure.

Stability looks at first like a defence: bootstrap ARI over 40 resamples is 0.977, 0.952, 0.932, 0.903 at $k=2, 3, 4, 5$ and falls away after (0.797, 0.581), so $k=4$ looks close to the largest reproducible k .

That reading does not survive a null calibration. Rebuilding the pipeline on structureless surrogates — a Gaussian matched to the real 30×30 facet covariance, and a copula matched to the covariance and every per-facet marginal — reproduces the stability number exactly: ARI 0.911 and 0.921 at silhouette 0.075 and 0.078, against the real data’s 0.910 at 0.082. A separated 4-component control confirms the test has power (silhouette 0.672, ARI 1.000).

At $n=410,168$ any smooth unimodal cloud with this covariance is partitioned reproducibly, so ARI ≈ 0.91 measures estimator variance, not multimodality, and neither criterion supports $k=4$. The clusters are not demographic artefacts either (Cramér’s V of 0.055–0.057 against sex, age band and country); they are a reproducible discretisation of a continuum, which bounds the cluster-conditional signal any persona can exploit. Since stability-based model selection is common, the general point is worth stating: at large n , bootstrap agreement must be read against a covariance-matched null or it certifies nothing.

D What the persona channel carries

A persona reaches the model only through `get_modifier_bisect`, which writes each score as one of five words on width-20 bins; each cluster’s genotype names five traits and five of the thirty facets, and every other sentence of the system prompt is shared. Ten words therefore carry the respondent’s identity, and two respondents sharing those ten bins would receive byte-identical prompts and so must receive identical beliefs.

We estimate what that channel permits. Within each cluster we fit $\mathbb{E}[X_j \mid \text{code}]$ on respondents *excluding* the forty personas, then evaluate on those forty, so the quantity is directly comparable to $\text{VR}_{\text{between}}$ and is never fitted on its own evaluation set.

Conditioning available to the model	Ceiling
Ten scores, five words each (<i>the audited system</i>)	0.482
same, with boosted trees	0.483
Ten scores, written as numbers	0.512
All 35 scores, five words each	0.707
All 35 scores, written as numbers	0.726
<i>best model observed</i>	0.295

Three things follow. The quantisation is not the binding constraint: writing numbers instead of words buys 0.03. The channel is not degenerate: all forty personas in all four clusters receive dis-

tinct codes, so no model is forced to answer two personas identically. And interactions add nothing (0.483 against 0.482), so the additive fit is essentially the conditional mean rather than a loose lower bound.

The oracle is fitted on this population and so knows its code-to-response mapping, which a zero-shot model does not; we therefore do not claim that a model *ought* to reach 0.482. The comparison rules out one explanation only — that the conditioning carries too little information to individuate — and that is the explanation the flatness of VR_{between} across eleven models most invites.

Where the shortfall falls. Each cluster names five facets, so twenty items have their own facet stated and one hundred are covered only at trait level. The named items carry $1.51\times$ more predictable-from-code signal (an SD ratio) (reference 0.671 against 0.444), and every model does individuate them more in absolute terms. Measured against those references the ratio is *lower* where the prompt is most specific — 0.353 on the named items against 0.433 on the rest, in the same direction for 10 of 11 models — but we report this descriptively and draw no causal conclusion from it, for two reasons. A facet score is computed from its own four items, so $\mathbb{E}[X_j \mid \text{facet}]$ is close to an arithmetic identity that a model without the scoring key cannot invert, while traits have the same structure diluted over twenty-four items; the named-item reference is inflated by more than the trait-level one, which alone could produce the gap. And the eleven models share personas, references and prompt structure, several within one family, so they are not independent replicates and we do not attach a p -value to the count.

Widening the channel does not fix it. The obvious remedy is to send more: we re-ran three models with every other word of the prompt held fixed and the ten modifier phrases replaced by the numbers themselves. The effect is inconsistent in magnitude — VR_{between} rises for Granite-4.1 (0.207 $\rightarrow$ 0.246) and gpt-oss (0.126 $\rightarrow$ 0.146) but falls for Apertus (0.130 $\rightarrow$ 0.096) — and also inconsistent in direction, the aligned amplitude moving 0.081 $\rightarrow$ 0.134 for Granite-4.1 and 0.039 $\rightarrow$ 0.047 for gpt-oss but 0.033 $\rightarrow$ 0.020 for Apertus. Granite gains on both counts — its correlation with the oracle rises from 0.285 to 0.453 — so for that model the richer encoding is a real improvement, while Apertus loses spread and direction together. On three models a

richer encoding is therefore a plausible but unreliable remedy, and we report it as a pilot rather than a recommendation.

The alignment does not depend on the oracle.

Because r is measured against a fitted target, an oracle missing structure the models track would understate it. It does not. Boosted trees on the same code change r by at most 0.012 and a ridge on the continuous scores by at most 0.034; an oracle given all thirty-five scores lowers r substantially, by up to 0.168, which is the expected direction since models only ever see the ten-score code. The ordering of models is unchanged throughout. Two independent computations agree where theory demands it: the ceiling as a standard-deviation ratio is 0.482 and the oracle’s correlation with actual respondents is 0.491, which coincide for a conditional mean. The check that assumes no oracle at all — correlating each model’s persona means directly against the forty respondents’ own answers — gives 0.076–0.248 against the oracle’s 0.491, so models recover between a sixth and a half of the attainable person-level alignment.

E Finite-sample behaviour of the anchors

Both anchors are estimated from $n=40$ per side, which for a classifier over 120 features is a small-sample regime. We sweep n on human-only data, where the true answers are known: two disjoint samples from the same cluster must be indistinguishable, so the honest C2ST value is exactly 0.5 and any excess is estimator bias.

n	20	40	80	160	320	640
C2ST human	—	0.574	0.542	0.536	0.527	0.517
DF human	0.915	0.939	0.957	0.969	0.978	0.985
C2ST const.	—	0.914	1.000	1.000	1.000	1.000

Two corrections follow, in opposite directions, and both land on their theoretical values. Fitting $y = y_\infty + c n^{-1/2}$ over the non-degenerate range gives an extrapolated C2ST human ceiling of **0.500** and an extrapolated DF human ceiling of **1.000** — exactly what two samples from the same distribution must score. So the C2ST ceiling of 0.574 at $n=40$ is *entirely* finite-sample classifier bias and should be read with its n attached, while the DF ceiling of 0.939 is *entirely* plug-in Wasserstein bias and is pessimistic. Correcting the latter widens the gap to the simulator’s 0.783 rather than narrowing it.

Below $n \approx 40$ *this* classifier stops separating even the constant from real humans (the dashed

entries above are the degenerate 0.5). That floor is a property of the configuration we used, not of C2ST: relaxing the minimum leaf size from 20 to 5 restores separation at $n=20$. We therefore recommend reporting every anchor together with the n and the classifier settings it was measured at, and treating any small- n floor as something to establish per configuration rather than to inherit from us.

Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. 2022. Large language models are human-level prompt engineers. In *International Conference on Learning Representations*.

References

- Yifan Dong and 1 others. 2025. Can large language model agents reliably express personality? *arXiv preprint*.
- Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. 2024. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In *International Conference on Learning Representations*.
- Shashank Gupta, Vaishnavi Shrivastava, Ameet Deshpande, Ashwin Kalyan, Peter Clark, Ashish Sabharwal, and Tushar Khot. 2024. Bias runs deep: Implicit reasoning biases in persona-assigned language models. In *International Conference on Learning Representations*.
- Guangyuan Jiang, Manjie Xu, Song-Chun Zhu, Wenjuan Han, Chi Zhang, and Yixin Zhu. 2024. Evaluating and inducing personality in pre-trained language models. In *Advances in Neural Information Processing Systems*.
- John A. Johnson. 2014. Measuring thirty facets of the five factor model with a 120-item public domain inventory: Development of the ipip-neo-120. *Journal of Research in Personality*, 51:78–89.
- David Lopez-Paz and Maxime Oquab. 2017. Revisiting classifier two-sample tests. In *International Conference on Learning Representations*.
- Joon Sung Park, Carolyn Q. Zou, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Robb Willer, Percy Liang, and Michael S. Bernstein. 2024. Generative agent simulations of 1,000 people. In *arXiv preprint arXiv:2411.10109*.
- Archiki Prasad, Peter Hase, Xiang Zhou, and Mohit Bansal. 2023. GrIPS: Gradient-free, edit-based instruction search for prompting large language models. In *Proceedings of the 17th Conference of the European Chapter of the ACL*.
- Greg Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. 2023. Personality traits in large language models. *arXiv preprint arXiv:2307.00184*.